

SHEN ZHENG

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EDUCATION

Carnegie Mellon University

PhD in Robotics | GPA: 4.06/4.33 | Advisor: [Dr. Srinivasa Narasimhan](#)

Aug. 2024 — Current

Pittsburgh, PA

Carnegie Mellon University

Master of Science in Computer Vision (MSCV) | GPA: 4.07/4.33 | Advisor: [Dr. Srinivasa Narasimhan](#)

Aug. 2022 — Dec. 2023

Pittsburgh, PA

- Coursework (A+): Learning-based Image Synthesis, Visual Learning and Recognition, Computer Vision, MultiModal Machine Learning, etc.

Wenzhou-Kean University

Bachelor of Arts in Mathematical Sciences | Major GPA: 3.94/4.00 | Advisor: [Dr. Gaurav Gupta](#)

Sep. 2017 – Jun. 2021

Wenzhou, China

- Honors: Dean's List (Top 3%); Zhejiang Provincial Government Scholarship (Top 3%); Outstanding Graduate (Top 10%).

PROFESSIONAL EXPERIENCES

Waymo

Summer 2025 & 2026

- Perception Software Engineer Intern: vectorized HD map construction with memory fusion (2025), 3D occupancy with Gaussian rendering (2026).

Lucid Motors

Feb. 2024 – Jul. 2024

- Perception Software Engineer in the ADAS perception team responsible for ADAS parking, traffic light detection, and blockage detection.

SELECTED PUBLICATIONS

- **Shen Zheng**, et al., *MapLightning: Online Vectorized HD Map Construction with 1D Map Tokens*. **Under Review**.
- **Shen Zheng**, et al., *WarpI2I: Image Warping for Image-to-Image Translation*. **ECCV 2026**.
- Anurag Ghosh, **Shen Zheng**, et al., *ROADWork Dataset: Learning to Recognize, Observe, Analyze and Drive Through Work Zones*. **ICCV 2025**.
- **Shen Zheng**, et al., *Instance-Warp: Saliency Guided Image Warping for Unsupervised Domain Adaptation*. **WACV 2025**.
- **Shen Zheng**, et al., *TPSeNCE: Towards Artifact-Free Realistic Rain Generation for Deraining and Object Detection in Rain*. **WACV 2024**.
- **Shen Zheng**, et al., *PointNorm: Dual Normalization is All You Need for Point Cloud Analysis*. **IJCNN 2023** (Oral).
- **Shen Zheng** and Gaurav Gupta, *Semantic-Guided Zero-Shot Learning for Low-Light Image/Video Enhancement*. **WACV 2022**.

RESEARCH EXPERIENCES

WarpI2I: Image Warping for Image-to-Image Translation [\[Webpage\]](#)

Jan. 2025 – Jan. 2026

- Proposed a warping framework that enlarge salient regions (e.g., objects, faces) to better preserve fine details in the compressed latent during I2I.
- Achieved model-agnostic gains across human relighting, driving relighting, and weather/time-of-day translation with only +0.006s latency.

ROADWork Dataset: Learning to Recognize, Observe, Analyze and Drive Through Work Zones [\[Webpage\]](#)

Feb. 2024 – Nov. 2024

- Proposed ROADWork, the largest open-source work zone dataset, to enhance baseline and foundation models in understanding work zones.
- Achieved +12.8× work zone discovery, +26.2% mAP for object detection, +36.7 SPICE for image captioning, and -21.3% error for path prediction.

Instance-Warp: Saliency Guided Image Warping for Unsupervised Domain Adaptation [\[Webpage\]](#)

May. 2023 – Jan. 2024

- Propose an instance-level image warping to oversample objects, enhancing object feature learning amidst dominant and highly varied backgrounds.
- Integrate image warping and feature unwarping into unsupervised domain adaptation in a task-agnostic way without test-time warping.
- Achieved +6.1% mAP50 for BDD100K Clear → DENSE Foggy object detection and +6.3% mIoU for Cityscapes → ACDC semantic segmentation.

TPSeNCE: Towards Artifact-Free Realistic Rain Generation for Deraining and Object Detection in Rain [\[Webpage\]](#)

Dec. 2022 – Apr. 2023

- Introduce a Triangular Probability Similarity (TPS) loss based on GAN discriminator to minimize the artifacts and distortions during rain generation.
- Propose a Semantic Noise Contrastive Estimation (SeNCE) strategy based on patch similarity and semantic similarity to optimize the rain amount.
- Obtained the best results for rain generation (KID=72.19), rain removal (MUSIQ=61.853), and object detection (mAP50=52.6%) on BDD100K.

PointNorm: Dual Normalization is All You Need for Point Cloud Analysis [\[Webpage\]](#)

Jun. 2022 – Aug. 2022

- Introduced PointNorm, a point cloud analysis network leveraging local mean and global standard deviation to boost loss and gradient stability.
- Proposed a plug-and-play DualNorm that normalize sampled and grouped points to each other to optimize the point cloud density.
- Achieved superior results on shape classification at ScanObjectNN (OA = 86.8%), part segmentation at ShapeNetPart (Inst. mIoU = 86.2%).

Semantic-Guided Zero-Shot Learning for Low-Light Image/Video Enhancement [\[Webpage\]](#)

Apr. 2021 – Jun. 2021

- Proposed a semantic-guided, zero-shot, low-light image enhancement network that consolidates high-level semantics into low-level enhancement.
- Constructed a recurrent image enhancement network that only demands an enhancement factor map with five non-reference loss functions.
- Attained the state-of-the-art UNIQUE/BRISQUE (0.805/27.01), mIoU/mPA (65.87%/74.50%) and inference time (0.001s).

ACADEMIC ACTIVITIES

- **Reviewer:** AAAI 2022, IJCNN (2023–2025), WACV (2023–2025), ECCV 2024, ICCV 2025, CVPR (2025–2026), BMVC 2026, TNNLS, IJCV.
- **Co-Instructor**, MATH 3291/3292: *Computer Vision*, Wenzhou-Kean University, Summer 2023 [\[Slides\]](#) [\[Recordings\]](#).
- **Guest Speaker:** *Image Processing with Machine Learning*, Fudan University, Spring 2021 [\[Slides\]](#).

SKILLS

- **Languages:** Python, C++, Java, R, Matlab, Shell, HTML, LaTeX, Markdown
- **Frameworks & Tools:** PyTorch, TensorFlow, Keras, MMCV, Docker, Git, ONNX, CUDA, Jupyter, Ubuntu
- **Libraries:** Scikit-Learn, NumPy, OpenCV, SciPy, Pandas, Matplotlib, Seaborn
- **DL Models:** CNN, RNN, GAN, VAE, Diffusion, Transformer